

Low pass Filter.

$$x(t) \xrightarrow{[R]} \xrightarrow{[C]} y(t) \text{ --- GND}$$

KV2.

$x(t)$ - input voltage.

$y(t)$ - output across cap

$R \rightarrow$ Resistance

$C \rightarrow$ Capacitance.

Voltage across capacitor.

$$x(t) - V_R(t) - y(t) = 0$$

$$x(t) - y(t) = V_R(t). \quad \text{--- (i)}$$

Current through capacitor.

$$i(t) = C \cdot \frac{dy(t)}{dt}$$

Current across cap & it is same.

$$x(t) - y(t) = R \cdot \frac{dy(t)}{dt}$$

$$x(t) = RC \frac{dy(t)}{dt} + y(t).$$

Now compare to original form.

$$x(t) = a_1 \frac{dy(t)}{dt} + a_0 y(t). \quad \text{--- (ii)}$$

where $a_0 = 1$, $a_1 = RC$.

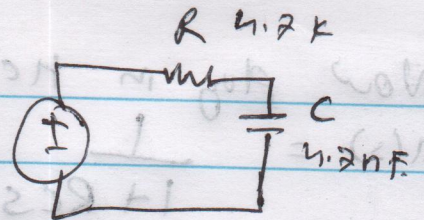
LHS $x(t)$ has no derivative $\Rightarrow x(t) \Rightarrow b_1 = 0$

From (ii)

$$a_0 y(t) + a_1 \frac{dy(t)}{dt} = b_0 x(t) + b_1 \frac{dx(t)}{dt}$$

Simple LPF digital

Say if cut off is $\omega_c = \frac{1}{RC} = \omega_c$



$$\omega_c = \frac{1}{RC} = \frac{1}{4.7 \times 10^3 \times 4.7 \times 10^{-9}} = \frac{1}{(4.7)^2 \times 10^{-6}}$$

$$\omega_c = \frac{1}{RC} \Rightarrow 1 = \frac{1}{2\pi f_c RC} \Rightarrow f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 4.7 \times 10^3 \times 4.7 \times 10^{-9}}$$

$$\Rightarrow \frac{1}{2\pi \times 4.7 \times 10^3 \times 4.7 \times 10^{-9}} = \frac{10^6}{2\pi \cdot (4.7)^2}$$

$$= 720.48 \text{ Hz}$$

$$f_c = 720.48 \text{ Hz}$$

$$H(s) = \frac{C \frac{dI}{dt}}{R I + C \frac{dI}{dt}}$$

$$H(s) = \frac{1}{\frac{R}{C} + \frac{1}{s}}$$

$$H(s) = \frac{s}{RCs + 1} = \frac{1}{1 + RCs} = H_G$$

Laplace to $[n]x = (s)x$ Laplace.

2 donan

$$\text{Substake } s = \frac{2}{T} \ln \frac{1-z^{-1}}{1+z^{-1}} \quad \text{where } T = \frac{1}{Fs}$$

$$\text{if } F_s = 44100 \text{ Hz} \Rightarrow T = 2.2676 \times 10^{-5} \text{ sec.}$$

$$\text{Now, } \frac{2}{44100} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 88200 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \quad \text{--- (11)}$$

Now plug in HCS

$$H(s) = \frac{1}{1 + RCs}$$

$$H(z) = \frac{1}{1 + (4.7)^2 \times 10^{-6} \times 88200 (1 - z^{-1}) (1 + z^{-1})}$$

$$= \frac{(1 + z^{-1})}{1 + 19.4834 (1 - z^{-1})}$$

$$= \frac{1 + z^{-1}}{1 + 19.4834 - 19.4834 z^{-1}}$$

$$= \frac{1 + z^{-1}}{20.4834 - 19.4834 z^{-1}}$$

Expressing in standard form

$$H(z) = \frac{b_0 + b_1 z^{-1}}{a_0 + a_1 z^{-1}} = \frac{X(z)}{Y(z)} \quad (14)$$

$$\Rightarrow b_0 = 1, b_1 = 1, a_0 = 20.4834, a_1 = -19.4834$$

$$X(z) (b_0 + b_1 z^{-1}) = Y(z) (a_0 + a_1 z^{-1})$$

$$X(z) b_0 + b_1 z^{-1} X(z) = Y(z) a_0 + Y(z) a_1 z^{-1} \quad (15)$$

Taking inverse Z-Transform, taking to time domain.

$$z^{-1} Y(z) \leftrightarrow y[n-1], \quad Y(z) = y[n]$$

$$z^{-1} X(z) \leftrightarrow x[n-1], \quad X(z) = x[n]$$

From (15)

$$x[n] b_0 + b_1 x[n-1] = a_0 y[n] + a_1 y[n-1]$$

$$y[n] = \frac{x[n] b_0 + b_1 x[n-1] - a_1 y[n-1]}{a_0} \quad (16)$$

First order LPF. Eqn. to make it easy make $a_0 = 1$